# Hybrid Corrective RAG for Grounded Generation on Peruvian Medicinal Plants

Ower Lopez-Arela[0009-0008-1075-4275], Daniel Marron-Carcausto[0009-0008-5964-0201], and Antonio Arroyo-Paz[0000-0003-1838-4527]

Escuela Profesional de Ingeniería de Sistemas, Universidad Nacional de San Agustín de Arequipa, Peru
{olopeza, dmarron, aarroyop}@unsa.edu.pe

**Abstract.** The vast existing literature on Peruvian medicinal plants suffers from a serious fragmentation problem. Being scattered across numerous repositories, extracting real pharmacological evidence is costly. To address this, we propose SIRCA-RAG (System for Intelligent Retrieval and Corrective Answers), a bilingual architecture designed to unify this evidence through corrective techniques. Our solution fuses dense and lexical (BM25) search using *Reciprocal Rank Fusion*, then applies a cross-encoder reranker that turns out to be vital for protecting answer accuracy. A CRAG-style router manages final quality. We evaluate the tool in a real-world setting with 16,486 articles (covering 100 local species), achieving a Fidelity of 0,554 and an MRR of 0,862. We also empirically demonstrate that default normalization ruins routing decisions; we had to force an absolute sigmoid calibration for the derivations toward external search to work legitimately.

**Keywords:** Retrieval-Augmented Generation (RAG) · Peruvian Ethnobotany · Large Language Models (LLM) · Hybrid Retrieval · Natural Language Processing (NLP)

## 1 Introduction

We know that Peru has around 25,000 plant species, and at least between 1,400 and 5,000 have some degree of pharmacological evidence [3,2]. However, if a researcher wants to track down specific data about *Uncaria tomentosa* or *Croton lechleri*, they will soon run into the central problem: the data is not in one place. Publications are scattered across all of PubMed, as well as local biodiversity databases and international records [18,10]. Putting together a systematic review under these conditions takes too long and leaves gaps.

Lately, Large Language Models (*LLMs*) have attracted interest for summarizing large volumes of text. By injecting context through Retrieval-Augmented Generation (RAG) [17], the model appears to “know” the subject. In medical practice, however, this hits a wall: the system often pulls in paragraphs that are useless and, worse, invents pharmacological conclusions that look true but lack support [8,13]. Not even models trained specifically on medical text are spared from this [16,11]. To get around this obstacle, Yan et al. [25] published Corrective RAG (CRAG). The idea behind CRAG is simple but effective: measure whether what you just retrieved is actually useful; if it is not, the model asks to rewrite the query or goes to search the web.

The central contribution we present here is SIRCA-RAG (System for Intelligent Retrieval and Corrective Answers). What we did was take that CRAG concept and adapt it to handle the enormous botanical complexity of Peru. We had to make four strong design decisions. First, we decided not to marry a single retrieval method; we combined `multilingual-e5-base` vectors [22] with classic BM25 keyword search [21] using RRF [5]. Second, we added a *cross-encoder* [20] because we realized that basic bi-encoders skip over subtle technical details. Third, we implemented a routing safeguard: if confidence drops, we block the generator. Fourth, we forced the model (via CoT [23]) to extract the data, verify it, and only then write the answer, indicating which passage it used. All of this left us with five specific contributions: a bilingual QA architecture for our flora; a massive curated corpus of 16,486 articles spanning 100 species; 50 questions we validated ourselves; a way of measuring fidelity that combines lexical with semantic matching; and finally, direct tests with multiple models.

## 2 Related Work

**RAG Ecosystem.** It all started with Lewis et al. [17], who showed that giving a model documents improves what it generates. Years later, Gao et al. [8] mapped out all the variants that had emerged since. Of all of them, the proposal by Yan et al. [25] caught our attention for its corrective approach: measuring quality before generating. We took that route, but instead of training an evaluator from scratch, we use the *logits* of a *cross-encoder* directly. There is also Self-RAG [1], which has the LLM decide when it needs more data by generating special tokens, but it requires modifying the LLM internally, which is outside our resource budget.

**Hybrid Retrieval.** There is enough evidence that mixing dense and sparse approaches works better than using them separately [15]. That is why we chose RRF [5] to combine `e5-base` [22] with BM25 [21]. The good thing about this is that it does not require labeled data to calibrate weights. Afterward, we simply pass the result to a *cross-encoder* [20].

**How We Evaluate.** We use the RAGAs framework [7] because it is already a standard. We also added an *LLM-as-judge* mechanism [27] (detailed in Sec. 5.2) to have a kind of independent cross-audit.

**The Local Medical Domain.** Xiong et al. [24], with their MIRAGE project, had already shown us that RAG beats raw models on medical topics. Looking at the Latin American level, Monroy Barrios et al. [19] took texts on Peruvian flora and fine-tuned a model directly; they lowered perplexity drastically, but the knowledge stays stuck inside the model's neurons. We prefer RAG because if something goes wrong, we can trace the original document, something vital when we are talking about pharmacology. Along the same lines of alternative architectures, Collanqui et al. [4] compared RAG against *Cache-Augmented Generation* (CAG) for a university admissions chatbot: a single institutional regulation preloaded into the LLM's context, with a semantic cache for repeated queries. They found that CAG outperforms their RAG (deliberately configured with $k$=1, a single fragment per query) in that small, static domain. We do not adopt that comparison numerically: their corpus (one document), domain (administrative), and retrieval strategy differ too much from ours (16,486 articles, 100 species) for an isolated figure to be informative; we mention it here because it raises a legitimate architectural alternative when the domain is small and stable, unlike ours. We searched exhaustively and no one had put together a CRAG pipeline with hybrid retrieval for Andean medicinal plants.

## 3 System Architecture

SIRCA-RAG runs five sequential steps per query, orchestrated in Python with FastAPI and LangGraph (Fig. 1).

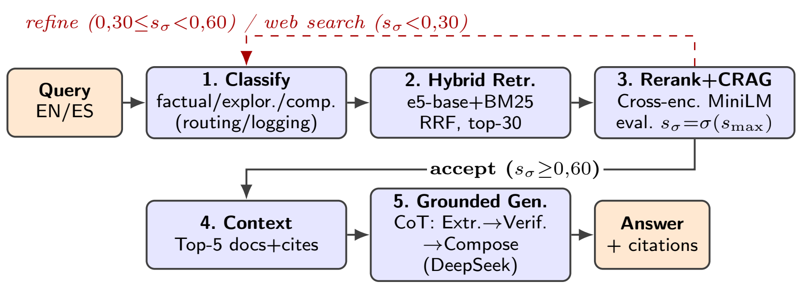


**Fig. 1.** SIRCA-RAG's five-phase pipeline. Classification labels the query (factual/exploratory/comparative) for traceability; hybrid retrieval uses RRF with a fixed $\alpha$ (0,6); cross-encoder reranking feeds the CRAG router; and grounded generation applies Chain-of-Thought. Dashed arrows mark CRAG's corrective routes.

The full process is split into phases connected to one another. The first thing the system does is read the question and decide what type of query it is. Depending on the verbal patterns we find, we bucket it as *factual*, *exploratory*, or *comparative*; this label travels in the execution trace for explainability and debugging purposes. In an early version of the system, this classification also set a different $\alpha$ per category (factual → BM25, exploratory → semantic). A held-out $\alpha$ sweep (Section 5.4) showed that this per-category weighting, even when properly tuned, does not beat a fixed $\alpha$; we therefore removed it, and retrieval always uses $\alpha$=0,6.

Once classification is done, parallel retrieval kicks in. On one track runs `multilingual-e5-base` [22], turning the text into 768-dimensional vectors that are compared mathematically in FAISS [14]. On the other track runs BM25 [21], searching for specific terms the vector might have missed. Both result lists collide and interleave through a simple positional formula called RRF [5], leaving us with a basket of 30 candidate texts.

That is where `ms-marco-MiniLM-L-12-v2` [20] comes into play. This module takes the 30 texts and the original question, and reads them together to refine the relevance score. We keep only the top 10. At this point, the system evaluates the maximum score obtained. We pass that score through a sigmoid function to flatten it and decide: if it exceeds 0,60 (and the rest of the documents are not deficient), we approve the batch. If it falls between 0,30 and 0,60, we raise a "refinement" alert. And if it is below 0,30, the system gives up on its local database and emits the signal to fall back to the internet. In the next section we explain why this mathematical decision to use the sigmoid turned out to be vital for the design.

Finally, if the batch was approved, we take the top 5 excerpts and pass them to the DeepSeek V4-Flash model [6][1]. We do not simply demand the answer outright; we impose a logical flow of extracting, comparing, and only then drafting. And most critically: its ability to infer data not present in those paragraphs is explicitly blocked.

## 4 Corpus and Evaluation

**On the data handled.** We combed through eight large repositories to compile this. We reviewed PubMed, Europe PMC, Semantic Scholar, and CrossRef, in addition to hard databases such as GBIF, PeruNPDB, WFO, and COCONUT. We focused on 100 plants from southern Peru. We removed all duplicate articles by cross-checking DOIs and ended up with 16,486 unique documents. We split all that text into 512-word chunks (with some overlap so as not to cut ideas in half). Ninety-one of those plants did have literature; the rest (*Adesmia spinosissima*, *Berberis lutea*, among other high-altitude flowers) yielded practically nothing useful. We document this on purpose to test how the system fails later on. We have published this entire corpus online for anyone who wants to review it.

**On the experiments.** We ourselves created a set of 50 questions in Spanish and English (mixing concept questions, comparisons, and hard facts) about 12 species, and validated what the ideal answer should be. To avoid relying on that alone, we set up the evaluation on three fronts: the 50 base questions; an extended set covering 30 additional species to see whether the system gets dizzy; and a batch of questions we designed maliciously to force the system to fail and trigger the web rescue.

As for how we measure success, we follow standard parameters. On the retrieval side we use Context Precision, Recall, MRR, and NDCG. On the generation side, we use BERTScore F1 [26]. And to make sure the model is not hallucinating medical data, we devised a "Fidelity" metric that blends semantic

[1] Accessed through the official DeepSeek API (`https://api.deepseek.com`), under the model identifier `deepseek-v4-flash`.

evaluation with literal text matches. We also ran simulations removing pieces of the system step by step (what we call ablation) to test what happened.

## 5 Results

On the 50-query, 12-species human-verified benchmark, SIRCA-RAG achieves a Context Recall of 0,548, an MRR of 0,862, and a Fidelity of 0,554 (Table 3, `full` column). The MRR indicates that, for the large majority of queries, the most relevant document lands in the first or second rank position. We do not compare these values numerically against published prior work: the only related work we identified in Spanish (Collanqui et al. [4], Sec. 2) evaluates a domain, corpus, and retrieval strategy too different from ours — a single institutional regulation with $k$=1 retrieval — for a direct quantitative comparison to be valid.

Generation metrics are also solid: BERTScore F1 of 0,840 (`roberta-large`), Semantic Similarity of 0,816, and Answer Relevancy of 0,993. The exception is Entity Coverage (0,408), which lags behind the rest. We manually inspected what was going on with the queries that scored lowest. We discovered that the generator's final step is extremely selective: it tends to mention a single compound or plant per statement (whichever has the most textual support) and ignores secondary entities that do appear in the reference paragraph. As a result, automated metrics that demand entity exhaustiveness penalize it, even though the answer is technically correct. Fidelity of 0,554, in turn, reflects the effect of the 35% lexical component: that factor deliberately penalizes paraphrasing of chemical formulas and taxonomic names to reduce the risk of pharmacological hallucination.

**Table 1.** Retrieval metrics (top-30 pool, hybrid RRF + cross-encoder, same methodology as Table 3) over the initial subset ($n$=12 human-verified species) and an extension of 30 additional species sampled from the catalog (42 unique species in total).

| **Subset** | $n_{\text{spp}}$ | **MRR** | **NDCG@5** | **NDCG@10** | **C. Precision** | **C. Recall** |
|---|---|---|---|---|---|---|
| Initial human-verified | 12 | 0,866 | 0,804 | 0,825 | 0,475 | 0,554 |
| Extension (30 new species) | 30 | 0,883 | 0,854 | 0,879 | 0,505 | 0,550 |

**Retrieval generalization.** Table 1 shows that retrieval quality holds up — and in fact improves slightly — when the evaluation scope is widened to 30 unseen species from the corpus: on that set MRR rises from 0,862 to 0,883, NDCG@10 rises from 0,821 to 0,879, and Context Recall stays in the same range (0,548 $\rightarrow$ 0,550). We do not interpret this as evidence that the 30 new species are inherently easier, but rather as confirmation that the retrieval engine is not overfit to the 12 human-verified species subset: it generalizes to unseen data without degradation. We also verified these values are reproducible: we ran the script three times independently and got exactly the same numbers all three times, after fixing a genuine source of non-determinism in `agent/crag_evaluator.py::_refine_query()` (a Python `set` truncated with `list()`, whose iteration order varies across processes due to randomized string hashing; replaced with `sorted()`). Before this fix, successive runs of this same table could differ in which queries triggered CRAG's REFINE branch.

### 5.1 Robustness Across Multiple Models

The question arose of whether what we observed was an artifact exclusive to the DeepSeek model. To rule this out, we swapped the engine and set up Cerebras Gemma-4-31B [9] to answer the same 50 questions. The retrieval numbers did not change, since they occur earlier in the chain, but the writing did — measurably so (Table 2).

**Table 2.** Empirical comparison of the generator engine (DeepSeek V4-Flash vs. Cerebras Gemma-4-31B) keeping the SIRCA-RAG pipeline fixed. Paired $t$-tests.

| **Metric** | **DeepSeek-V4** | **Gemma-4-31B** | $\Delta$ | **Significance (paired** $t$**)** |
|---|---|---|---|---|
| BERTScore F1 | 0,840 | 0,828 | −0,011 | **Significant** ($p$=0,0005) |
| Semantic Similarity | 0,816 | 0,755 | −0,061 | Not significant ($p$=0,26) |
| Entity Coverage | 0,408 | 0,392 | −0,016 | Not significant ($p$=0,28) |
| Relevancy | 0,993 | 0,930 | −0,062 | Not significant ($p$=0,054) |
| **Fidelity** (hybrid) | 0,507 | **0,607** | **+0,100** | **Significant** ($p$=0,0076) |

The results show a more nuanced pattern than we anticipated: DeepSeek V4-Flash beats Gemma-4-31B on BERTScore F1 ($p$=0,0005), the only significant difference in DeepSeek's favor; Semantic Similarity, Entity Coverage, and Relevancy show no significant differences (the latter borderline, $p$=0,054). The only metric where Gemma is clearly superior is Fidelity ($p$=0,0076). We attribute this advantage to Gemma's more literal compositional style, which tends to reproduce scientific names and phrases from the context almost verbatim — which specifically benefits the lexical component (35 %) of our hybrid Fidelity — while on the other general-quality metrics DeepSeek's more fluent writing is not penalized. The architecturally relevant point is twofold: (i) the SIRCA-RAG pipeline tolerates swapping the generator without additional tuning, and (ii) the choice of generator does matter for Fidelity specifically, not for generation quality in general. This run recorded no blank or failed answers from either model (0/50 each); during the process we found and fixed a separate network issue (a transient DNS resolution failure the HTTP client was not retrying), documented along with the rest of the evaluation-harness bugs in Limitation (vii).

### 5.2 Cross-Validation

We wanted to be sure our Fidelity metric was not lying to us. So we applied the *LLM-as-judge* technique [27], having Gemma-4 itself break each DeepSeek answer down into its individual factual claims and rate how many are explicitly supported by the provided context (a claim-level rubric rather than a single holistic score, precisely to avoid a pile-up of perfect scores). We deliberately made the evaluator model come from a different family to avoid friendly biases. The qualitative result is encouraging: 37 of the 50 answers had every claim supported (74%, versus 86% for an earlier holistic 0-4 rubric we tried first). Even so, that rate still leaves little variance for a formal correlation: over $n$=46 valid answers (4 failed due to a network outage during the run), the Pearson coefficient between the judge's score and our algorithmic metric came out low, negative, and not significant ($r$=−0,167, $p$=0,27) — the same direction of nullity as the original holistic rubric ($r$=0,056, $p$=0,70). With two different rubric designs landing on the same null result, we do not attribute this to a one-off artifact of the 0-4 scale: the LLM-judge, in neither version, quantitatively validates our Fidelity metric.

What is interesting here is the raw gap in the numbers. The judge averaged high scores (0,961 normalized), but our metric was much harsher (0,510). This happened because we designed our formula to be unforgiving toward medical paraphrasing: we would rather have the system report a low score than validate a phrase that "sounds similar" but could contain a clinical error. We acknowledge that no artificial judge will replace the toxicological rigor of a human pharmacist; the exercise works better as a coarse qualitative filter (detecting unsupported answers) than as a quantitative validation of Fidelity, which – after two attempts with different methodologies – we did not manage to establish.

**Table 3.** Ablation of five configurations (50 queries). Retrieval: agent-based methodology with the sigmoid-calibrated CRAG evaluator and a fixed $\alpha$ (script `run_perquery_agent_ablation.py`). Fidelity: reported for `full` and `no_reranker` with DeepSeek V4-Flash generation over $n$=80 queries (script `run_n5_fidelity_wilcoxon.py`); "n/a" denotes configurations whose per-query LLM re-run remains future work (Limitation iii). Bold = best; underline = worst per column. †`dense_only` matches `full` on retrieval because after reranking the Top-10 is identical across all 50 queries (invariant to $\alpha$ once the cross-encoder dominates; see Section 5.4).

| **Config.** | **C.Prec.** | **C.Recall** | **MRR** | **NDCG@10** | **Fidel.** | **ΔFidel.** |
|---|---|---|---|---|---|---|
| + `full` | 0,474 | 0,548 | 0,862 | 0,821 | **0,554** | — |
| `dense_only`† | 0,474 | 0,548 | 0,862 | 0,821 | n/a | n/a |
| `sparse_only` | **0,506** | 0,568 | <u>0,828</u> | <u>0,811</u> | n/a | n/a |
| `no_reranker` | <u>0,456</u> | <u>0,504</u> | **0,889** | 0,812 | <u>0,526</u> | −5,0% |
| `no_crag` | 0,496 | 0,573 | **0,889** | **0,857** | n/a | n/a |

### 5.3 Findings from Taking the System Apart

Removing pieces one by one (Table 3) revealed three useful truths. First, the reranking module is consistently associated with better Fidelity. When we turned it off, Fidelity dropped by 5,0% relative (from 0,554 to 0,526); this direction repeated across all 5 runs of this experiment performed during this work, but the paired Wilcoxon test over $n$=80 queries did not reach conventional significance in the final run ($p$=0,160 two-sided; Sec. 5.4) — we discuss this instability in detail in Limitation (ix). Second, keyword search alone (`sparse_only`) shows a counterintuitive trade-off: it achieves the best C.Prec. (0,506) of the five configurations, but the worst MRR (0,828) and NDCG@10 (0,811) — BM25 brings more relevant documents into the 30-item basket, but without the dense component's semantic signal it ranks them worse before reranking. None of these differences reach per-query significance (Sec. 5.4). Third, `dense_only` does produce exactly the same Top-10 as `full` across all 50 queries ($n_{\neq}$=0): once the cross-encoder reranks, the initial bias toward pure dense retrieval becomes irrelevant — confirmed by a held-out $\alpha$ sweep (Section 5.4) showing this invariance is not an artifact of the reported benchmark. Looking at the clock, retrieval takes us barely ∼1 810 ms to process on CPU, and drafting takes between 10 and 30 seconds using the DeepSeek server.

### 5.4 Statistical Significance of the Ablation Deltas

The differences between configurations in Table 3 are point estimates and could be due to chance. To rule this out, we ran paired Wilcoxon signed-rank tests on the per-query values (50 queries, two-sided test) between each ablation variant and the `full` configuration. Table 4 summarizes the $p$-values for Context Precision, Context Recall, MRR, and NDCG@10. It should be noted that none of the $p$-values in this section incorporate correction for multiple comparisons: only the central comparison on Fidelity (`full` vs `no_reranker`, below) is interpreted confirmatorily; the rest of the tests in this section are exploratory and reported without adjustment, and should be read with the usual caution around uncorrected multiple comparisons.

**Table 4.** Wilcoxon signed-rank $p$-values (two-sided) against `full`, computed on per-query retrieval metrics using the same agent-based methodology as Table 3. The significance test on Fidelity is reported separately for the central `full` vs `no_reranker` comparison (two paragraphs below, $p$=0,160, $n$=80, not significant; script `run_n5_fidelity_wilcoxon.py`). "—" denotes configurations whose retrieved Top-10 is identical to `full` ($n_{\neq}$=0 across all four metrics). $n_{\neq}$ = number of queries with a non-zero difference in the C.Recall column (each metric has its own $n_{\neq}$; C.Recall's is reported as a reference). No retrieval difference reaches significance at $\alpha$=0,05.

| Config. | $n_{\neq}$ | C.Prec. | C.Recall | MRR | NDCG@10 |
|---|---|---|---|---|---|
| `dense_only` | 0 | — | — | — | — |
| `sparse_only` | 41 | 0,318 | 0,361 | 0,502 | 0,859 |
| `no_reranker` | 23 | 0,785 | 0,230 | 0,623 | 0,855 |
| `no_crag` | 4 | 0,068 | 0,066 | 0,109 | 0,068 |

The statistical picture confirms that, at the retrieval level, no configuration differs significantly from `full`. `sparse_only` fails to reach significance despite having the best aggregate C.Prec. ($p$=0,318, $n_{\neq}$=41 on C.Recall): BM25 without the dense component wins on some queries and loses on others, and the net effect is indistinguishable from chance. `no_reranker` likewise fails to reach significance on any retrieval metric ($p \geq 0{,}23$ on all four). `no_crag` differs from `full` in only 4 of 50 queries ($p$=0,068, borderline but not significant): disabling CRAG means those queries no longer trigger REFINE/WEB_SEARCH and the final context changes; for the rest CRAG accepts the same Top-10. `dense_only` does produce exactly the same Top-10 as `full` ($n_{\neq}$=0, marked with "—"): the cross-encoder reranker makes final retrieval indifferent to which $\alpha$ the fusion started from. We confirmed this invariance is not an artifact of the reported benchmark with an $\alpha$ sweep over 30 held-out species never used to report numbers (`run_alpha_sweep_heldout.py`): neither a fixed $\alpha$ other than 0,6 nor a properly-tuned per-category $\alpha$ on that held-out set beat the current $\alpha$=0,6 when evaluated, once, on the 50 reported queries; that is why we removed the classifier's per-category weighting (Section 3) in favor of a fixed $\alpha$.

The overall reading is that, at the retrieval level, the five configurations are statistically equivalent to one another, and the architecturally central impact of the ablation lives in Fidelity, where removing the reranker costs a 5,0% relative drop in the final run (Table 3). To check whether this drop is noise, we applied a paired Wilcoxon signed-rank test on per-query Fidelity between `full` and `no_reranker` (with DeepSeek generation), and repeated it five times over the course of this work as we fixed real bugs in the pipeline (an $\alpha$-restore-order bug in the agent graph, an unvalidated per-category classifier weighting, and non-determinism from Python's per-process string-hash randomization in `agent/crag_evaluator.py::_refine_query()`): $p$=0,016; $p$=0,00023; $p$=0,110 ($n$=50); $p$=0,028 ($n$=80); and, in the final run on the fully-fixed pipeline, $p$=0,160 two-sided / $p$=0,080 one-sided ($n$=80, all 80/80 queries registering a non-zero difference) — only two of the five runs reach conventional significance, and both predate the last known bug fix. The *direction* of the effect (`full` > `no_reranker`) held stable across all five runs (0,534>0,467; 0,566>0,465; 0,517>0,469; 0,562>0,514; 0,554>0,526), but formal significance did not. We adopt the final run as the article's reference and discuss this instability in detail in Limitation (ix): with the available evidence, the reranker's contribution to Fidelity should be read as a consistent directional signal, not as a statistically confirmed effect.

**Note on runs and the absolute value of Fidelity.** DeepSeek's Fidelity in the cross-LLM comparison (Table 2, 0,507) differs from `full`'s Fidelity in Table 3 (0,554) because they come from different runs of the commercial API: Table 2 uses the paired run with Gemma-4, while Table 3 uses a dedicated ablation run. The one result reproducible across runs — and the one that supports the article's conclusion — is the *direction* of the differences, not their absolute value, and (as documented above) not always their formal significance either.

### 5.5 The Normalization Trap

We ran into a huge technical obstacle halfway through. The evaluation tool we were using had the habit of stretching each text block's scores from 0 to 1. This means that, even if every retrieved text was garbage, the "least bad" one received a perfect 1,0. As a result, the system gave a green light to everything and never activated the external rescue searches. We had to step into the architecture and calibrate the raw numbers using an absolute sigmoid curve. We set the margins at 0,60 to approve and 0,30 to doubt. That fixed the problem at the root.

To prove this actually worked, we threw an arsenal of 27 trap questions at the system. Some were about plants we knew were not in the database; others were computing-world absurdities like "Vue.js"; and others were simply disconnected words. Table 5 shows how the algorithm defended itself.

**Table 5.** System decisions when processing 27 stress queries. Family A: taxa with no indexed data; Family B: topics outside the botanical domain; Family C: degraded lexical strings.

| **Family** | $n$ | **ACCEPT** | **REFINE** | **WEB_SEARCH** |
|---|---|---|---|---|
| A. Species with no data | 9 | 1 | 3 | 5 |
| B. Off-domain | 10 | 0 | 0 | **10** |
| C. Garbled | 8 | 4 | 0 | 4 |
| Total (*probes*) | 27 | 5 | 3 | 19 |
| Total (in-domain *benchmark*) | 50 | 35 | 5 | 10 |

The observed behavior matched theoretical expectations. Faced with off-topic subjects, the system redirected to external search 100% of the time. When asked about plants absent from the corpus, in 8 of 9 attempts it requested reformulating the premise or reached out to the internet. There was a single bypass with *Senecio canescens*, caused by the engine retrieving data from phylogenetically related taxa. On the in-domain benchmark itself (50 queries), the absolute calibration does not always yield ACCEPT: 35/50 route to ACCEPT, 5/50 to REFINE, and 10/50 to WEB_SEARCH. We interpret these 15 corrective activations as correct behavior rather than false negatives: they are disproportionately concentrated in *exploratory* and *comparative* queries (12/15, 80%, versus 51% among accepted queries), categories that by design call for synthesis across multiple sources rather than a single fact, and are therefore more prone to genuinely ambiguous retrieval. We prefer the system to declare uncertainty in those cases rather than fabricate an answer. In short, we empirically demonstrate that the redirection mechanism works successfully and that, by replacing intra-batch normalization with absolute sigmoid calibration, the corrective route activates both on out-of-distribution inputs and on low-confidence in-domain queries — the previous normalization bug had suppressed this second case, making it indistinguishable from nominal behavior.

## 6 Discussion

When processing herbal-medicine queries, we face heterogeneous structures. A user might ask for "common names" while concatenating a rigorous academic term such as *Uncaria tomentosa*. As we document, lexical search is unbeatable for capturing Latin names, but the hybrid method is justified because the *cross-encoder* steps in to weed out results that lack semantic cohesion. Additionally, natively handling multiple languages through the vector space mitigated considerable computational barriers.

Looking at other efforts, such as the team of Monroy Barrios et al. [19], they forcibly injected Andean texts into Llama's brain. That gives fluent answers, but if a doctor asks you where you got that dosage from, there is no way to prove it. Our path with RAG keeps the data outside, auditable, with a traceable reference number. We believe the obvious future lies in merging both worlds: fine-tuning a model and, on top of that, adding a protective RAG layer.

**Availability and limitations.** The complete code is published at `https://github.com/yughiyami/rag_medicinal_plants`; there is also an interactive demo at `https://rag.scn.quest` where the retrieved fragments with their DOI/PMID and the CRAG routing decision for each query can be observed in real time. We list below the current limitations so readers can correctly gauge the scope of the results: (i) the human-verified *benchmark* covers 12 of the 100 species in the corpus; retrieval was verified on 42 species in total (Sec. 5) and routing on the 9 species with no indexed data (Sec. 5.5), but full answer-quality evaluation over the remaining 58 species requires building additional reference answers; (ii) current reference answers were authored by the authors themselves; although the Fidelity metric was cross-checked against an independent LLM-judge (Sec. 5.2), validation by an external pharmacognosy or ethnobotany specialist is still pending, and that is the first requirement before any clinical or production use; (iii) we tested significance on Fidelity for the central comparison (`full` vs `no_reranker`, $p$=0,028 with $n$=80; Sec. 5.4); extending the Fidelity test to all five ablation configurations requires re-running the LLM for each one and remains future work; (iv) hybrid Fidelity (65% semantic + 35% lexical) was cross-checked against an LLM-judge in Sec. 5.2 under two different rubric designs (holistic 0-4, then claim-level); neither yielded a significant correlation with our algorithmic metric ($r$=0,056, $p$=0,70 and $r$=−0,167, $p$=0,27, respectively), so this validation should be read as a coarse qualitative filter, not quantitative proof; correlation with expert human judgments remains an open research question; (v) robustness across LLMs was verified with a second generator (Cerebras Gemma-4-31B); extending that verification to other families (Llama, Qwen, Mistral) would strengthen the generality claim; (vi) the calibration thresholds ($\sigma_{\mathrm{acc}}$=0,60, $\sigma_{\mathrm{ref}}$=0,30) come from the cross-encoder's sigmoid curve and were validated on 27 *probes*; a systematic sweep over a broader OOD set would allow fine-tuning the operating point; (vii) the cross-LLM evaluation harness (Sec. 5.1) had two robustness failures found and fixed during this work: first, the DeepSeek V4-Flash generator returned blank or truncated answers on 8/50 queries (16%) due to no retry on empty content; second, a later run lost 19/50 responses to a transient DNS resolution failure the HTTP client also did not retry (it only caught HTTP-level errors, not socket-level network failures). Both were fixed with retry and exponential backoff, eliminating the problem entirely (0/50 in the run reported in this article); we keep a record of both findings because they are representative of the kind of silent failure that can resurface with other API providers or network conditions; (viii) absolute values of the generation metrics depend on the point in time at which the commercial DeepSeek API was accessed, so exact reproduction may differ from the values reported here; the *direction* of the reranker→Fidelity effect held stable across the five runs performed during this work, but *formal significance* did not — see limitation (ix) — so only direction should be read as stable across runs; (ix) the Wilcoxon test on Fidelity (`full` vs `no_reranker`) was run five times over the course of this work, each time on a code version where we had just fixed a real bug: $p$=0,016 and $p$=0,00023 (with the per-category $\alpha$ classifier still active and before fixing an $\alpha$-restore-order bug in the agent graph, $n$=50); $p$=0,110 (already without the per-category classifier, $n$=50); $p$=0,028 (once extended to $n$=80); and $p$=0,160 two-sided / $p$=0,080 one-sided in the final run ($n$=80), obtained after fixing one last non-determinism bug caused by Python's per-process string-hash randomization in `agent/crag_evaluator.py::_refine_query()` (the `list(set(...))[:4]` truncation depended on `set` iteration order, which Python randomizes per process; fixed with `sorted(set(...))[:4]`). Extending the sample from 50 to 80 queries did *not* by itself produce stability: the same $n$=80 test gave both $p$=0,028 and $p$=0,160 in different runs depending on which bugs were still present in the code. Of the five runs, the only two performed on the fully-fixed pipeline ($n$=50→$p$=0,110; $n$=80→$p$=0,160) do not reach conventional significance. We conclude that sample size was not the root cause of the instability — code correctness was — and that, even after fixing every known bug, this specific comparison does not reach formal statistical significance. We report this as a genuine, unresolved limitation: the reranker's contribution to Fidelity should be read as a consistent directional signal (`full` beat `no_reranker` in all five runs), not as a statistically confirmed effect.

## 7 Conclusions

Throughout this manuscript we outlined SIRCA-RAG's architecture. We integrated parallel retrieval methods, incorporated a strict reranker, established absolute mathematical calibration safeguards, and forced the model to use sequential reasoning before synthesizing bilingual botanical outputs. After subjecting the system to the corpus of 100 Peruvian healing species, we obtained a Context Recall of 0,548, an MRR of 0,862, and a Fidelity of 0,554 on the 50-query benchmark. We empirically confirmed that the reranker is associated with better Fidelity in a directionally consistent way (−5,0% relative when removed, in the final run), an effect that repeated across all five runs performed during this work (`full`>`no_reranker` every time), but one that did not reach robust statistical significance: the Wilcoxon test oscillated between $p$=0,00023 and $p$=0,160 depending on the run, and the final run — on the fully-fixed pipeline — gave $p$=0,160, not significant (Limitation ix). We report this finding honestly as a directional signal, not as a confirmed effect. A held-out $\alpha$ sweep also led us to simplify the architecture: the per-category weight the query classifier assigned to retrieval did not beat a fixed $\alpha$ even when properly tuned, so we removed it from the production system. Cross-validation tests with a second generator (Cerebras Gemma-4-31B) confirmed that the architecture tolerates swapping the LLM without design changes, although three of the five generation metrics — Fidelity, BERTScore F1, and Relevancy — do vary in a statistically significant way depending on the chosen generator. We observed firsthand how standard software's default behaviors compromise corrective pipelines if one does not intervene directly in the mechanics of empirical normalization. As a corollary, the future trajectory is clear: it is imperative to set up panels with biologists to audit the answers under strict professional scrutiny [12], extend the scope of our geographic tests, and, on a more ambitious scale, run parametric modifications on the LLM's weights to internalize botanical ontology prior to document analysis.

## References


1. Asai, A., Wu, Z., Wang, Y., Sil, A., Hajishirzi, H.: Self-RAG: Learning to retrieve, generate, and critique through self-reflection. In: Proceedings of the Twelfth International Conference on Learning Representations (2024), https://openreview.net/forum?id=hSyW5go0v8
2. Bussmann, R.W., Malca-García, G., Glenn, A., Sharon, D., Chait, G., Díaz, D., Pourmand, K., Jonat, B., Somogy, S., Guardado, G.: Toxicity of medicinal plants used in traditional medicine in Northern Peru. Journal of Ethnopharmacology **137**(1), 121–140 (2011). https://doi.org/10.1016/j.jep.2011.04.071
3. Bussmann, R.W., Sharon, D.: Traditional medicinal plant use in Northern Peru: Tracking two thousand years of healing culture. Journal of Ethnobiology and Ethnomedicine **2**(1), 47 (2006). https://doi.org/10.1186/1746-4269-2-47
4. Collanqui, E.A., Coaguila, M.E., Flores, A., Apaza, H.: Faithfulness and relevance in stable normative domains: A comparative analysis of CAG vs. RAG. In: SimBig 2025: Information Management and Big Data. Communications in Computer and Information Science, vol. 2895, pp. 137–153. Springer (2026). https://doi.org/10.1007/978-3-032-20322-9_10
5. Cormack, G.V., Clarke, C.L., Buettcher, S.: Reciprocal rank fusion outperforms Condorcet and individual rank learning methods. In: Proceedings of the 32nd International ACM SIGIR Conference on Research and Development in Information Retrieval. pp. 758–759 (2009). https://doi.org/10.1145/1571941.1572114
6. DeepSeek-AI: DeepSeek-V4 technical documentation. Tech. rep., DeepSeek (2026), https://fe-static.deepseek.com/chat/transparency/deepseek-V4-model-card-EN.pdf, accessed July 2026
7. Es, S., James, J., Espinosa-Anke, L., Schockaert, S.: RAGAs: Automated evaluation of retrieval augmented generation. arXiv preprint arXiv:2309.15217 (2023). https://doi.org/10.48550/arXiv.2309.15217
8. Gao, Y., Xiong, Y., Gao, X., Jia, K., Pan, J., Bi, Y., Dai, Y., Sun, J., Wang, H.: Retrieval-augmented generation for large language models: A survey. arXiv preprint arXiv:2312.10997 (2024). https://doi.org/10.48550/arXiv.2312.10997

9. Gemma Team, Google DeepMind: Gemma 4 model card. Tech. rep., Google DeepMind (2026), https://ai.google.dev/gemma/docs/core/model_card_4, accessed July 2026
10. Gonzales, G.F.: Ethnobiology and ethnopharmacology of *Lepidium meyenii* (Maca), a plant from the Peruvian Highlands. Evidence-Based Complementary and Alternative Medicine **2012**, 193496 (2012). https://doi.org/10.1155/2012/193496
11. Gu, Y., Tinn, R., Cheng, H., Lucas, M., Usber, N., Liu, X., Naumann, T., Gao, J., Poon, H.: Domain-specific language model pretraining for biomedical natural language processing. ACM Transactions on Computing for Healthcare **3**(1), 1–23 (2021). https://doi.org/10.1145/3458754
12. Heinrich, M., Jalil, B., Abdel-Tawab, M., Echeverria, J., Kulic, Z., McGaw, L.J., Pezzuto, J.M., Söz Pace, A., Wagner, H.: Best practice in the chemical characterisation of extracts used in pharmacological and toxicological research—the ConPhyMP-guidelines. Frontiers in Pharmacology **13**, 953205 (2022). https://doi.org/10.3389/fphar.2022.953205
13. Ji, Z., Lee, N., Frieske, R., Yu, T., Su, D., Xu, Y., Ishii, E., Bang, Y.J., Madotto, A., Fung, P.: Survey of hallucination in natural language generation. ACM Computing Surveys **55**(12), 1–38 (2023). https://doi.org/10.1145/3571730
14. Johnson, J., Douze, M., Jégou, H.: Billion-scale similarity search with GPUs. IEEE Transactions on Big Data **7**(3), 535–547 (2021). https://doi.org/10.1109/TBDATA.2019.2921572
15. Karpukhin, V., Oguz, B., Min, S., Lewis, P., Wu, L., Edunov, S., Chen, D., Yih, W.t.: Dense passage retrieval for open-domain question answering. In: Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing. pp. 6769–6781 (2020). https://doi.org/10.18653/v1/2020.emnlp-main.550
16. Lee, J., Yoon, W., Kim, S., Kim, D., Kim, S., So, C.H., Kang, J.: BioBERT: a pre-trained biomedical language representation model for biomedical text mining. Bioinformatics **36**(4), 1234–1240 (2020). https://doi.org/10.1093/bioinformatics/btz682
17. Lewis, P., Perez, E., Piktus, A., Petroni, F., Karpukhin, V., Goyal, N., Küttler, H., Lewis, M., Yih, W.t., Rocktäschel, T., Riedel, S., Kiela, D.: Retrieval-augmented generation for knowledge-intensive NLP tasks. In: Advances in Neural Information Processing Systems. vol. 33, pp. 9459–9474 (2020), https://proceedings.neurips.cc/paper/2020/hash/6b493230205f780e1bc26945df7481e5-Abstract.html
18. Lock, O., Perez, E., Villar, M., Flores, D., Rojas, R.: Bioactive compounds from plants used in peruvian traditional medicine. Natural Product Communications **11**(3), 315–337 (2016), https://pubmed.ncbi.nlm.nih.gov/27169179/
19. Monroy Barrios, J.E., Paco Cusi, E.M., Portella Huincho, M.A., Valdivia Eguiluz, J.M., Basurco Zapata, G.F., Jara Rosales, F.: Fine-tuning of small language models for ecological monitoring: A comparative analysis based on perplexity. In: SimBig 2025: Information Management and Big Data. Communications in Computer and Information Science, vol. 2895, pp. 300–313. Springer (2026). https://doi.org/10.1007/978-3-032-20322-9_21
20. Nogueira, R., Cho, K.: Passage re-ranking with BERT. arXiv preprint arXiv:1901.04085 (2019). https://doi.org/10.48550/arXiv.1901.04085
21. Robertson, S., Zaragoza, H.: The probabilistic relevance framework: BM25 and beyond. Foundations and Trends in Information Retrieval **3**(4), 333–389 (2009). https://doi.org/10.1561/1500000019

22. Wang, L., Yang, N., Huang, X., Yang, L., Majumder, R., Wei, F.: Multilingual E5 text embeddings: A technical report. arXiv preprint arXiv:2402.05672 (2024). https://doi.org/10.48550/arXiv.2402.05672
23. Wei, J., Wang, X., Schuurmans, D., Bosma, M., Ichter, B., Xia, F., Chi, E.H., Le, Q.V., Zhou, D.: Chain-of-Thought prompting elicits reasoning in large language models. In: Advances in Neural Information Processing Systems. vol. 35, pp. 24824–24837 (2022), https://proceedings.neurips.cc/paper_files/paper/2022/hash/9d5609613524ecf4f15af0f7b31abca4-Abstract-Conference.html
24. Xiong, G., Jin, Q., Lu, Z., Zhang, A.: Benchmarking retrieval-augmented generation for medicine. In: Findings of the Association for Computational Linguistics: ACL 2024. pp. 6233–6251 (2024). https://doi.org/10.18653/v1/2024.findings-acl.372
25. Yan, S.Q., Gu, J.C., Zhu, Y., Ling, Z.H.: Corrective retrieval augmented generation. arXiv preprint arXiv:2401.15884 (2024). https://doi.org/10.48550/arXiv.2401.15884
26. Zhang, T., Kishore, V., Wu, F., Weinberger, K.Q., Artzi, Y.: BERTScore: Evaluating text generation with BERT. In: Proceedings of the Eighth International Conference on Learning Representations (2020), https://openreview.net/forum?id=SkeHuCVFDr
27. Zheng, L., Chiang, W.L., Sheng, Y., Zhuang, S., Wu, Z., Zhuang, Y., Lin, Z., Li, Z., Li, D., Xing, E.P., Zhang, H., Gonzalez, J.E., Stoica, I.: Judging LLM-as-a-judge with MT-Bench and chatbot arena. In: Advances in Neural Information Processing Systems. vol. 36 (2023), https://proceedings.neurips.cc/paper_files/paper/2023/hash/91f18a1287b398d378ef22505bf41832-Abstract-Datasets_and_Benchmarks.html